

1

A particle moves in a straight line so that its displacement (in m) from a fixed point O at time t seconds is given by $x = 2 \sin 2t$, $0 \leq t \leq 2\pi$.

Find:

- | | | |
|------|--|---|
| i. | The initial velocity | 1 |
| ii. | The acceleration after $\frac{\pi}{12}$ seconds. | 1 |
| iii. | When the particle is at rest. | 2 |
| iv. | The displacement of the particle when it is at rest. | 2 |

(a) $x = 2 \sin 2t$
 $\dot{x} = 4 \cos 2t$
 $\ddot{x} = -8 \sin 2t$

i. $t = 0 \quad \dot{x} = 4 \cos 2(0)$
 $= 4 \times 1$
 $= 4 \text{ m/s}$

ii. $t = \frac{\pi}{12} \quad \ddot{x} = -8 \sin 2\left(\frac{\pi}{12}\right)$
 $= -8 \sin\left(\frac{\pi}{6}\right)$
 $= -8 \times \frac{1}{2}$
 $= -4 \text{ m/s}^2$

iii. $\dot{x} = 0 \quad \text{then } 4 \cos 2t = 0$
i.e.
 $\cos 2t = 0$
 $2t = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \dots$
 $t = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}, \dots$

iv. $x = 2 \sin 2t$
 $= 2 \sin 2\left(\frac{\pi}{4}\right)$
 $= 2$

$\therefore x = \pm 2 \text{ m}$

2

The rate at which Carbon Dioxide will be produced when conducting an experiment is given by $\frac{dV}{dt} = \frac{1}{100}(30t - t^2)$ where $V \text{ cm}^3$ is the volume of gas produced after t minutes.

- i. At what rate is the gas being produced 15 minutes after the experiment begins. 1
- ii. How much Carbon Dioxide has been produced during this time? 2

(c) i. $\frac{dV}{dt} = \frac{1}{100}(30t - t^2)$

when $t = 15$

$$\begin{aligned}\frac{dV}{dt} &= \frac{1}{100} [30(15) - (15)^2] \\ &= \frac{225}{100} \\ &= 2\frac{1}{4} \text{ cm}^3 / \text{min}\end{aligned}$$

ii. $V = \int_0^{15} \frac{1}{100}(30t - t^2) dt$

$$\begin{aligned}&= \frac{1}{100} \left[15t^2 - \frac{t^3}{3} \right]_0^{15} \\ &= \frac{1}{100} \left\{ \left[15(15)^2 - \frac{15^3}{3} \right] - [0] \right\} \\ &= \frac{1}{100} [3375 - 1125] \\ &= \frac{1}{100} (2250) \\ &= 22.5 \text{ cm}^3\end{aligned}$$

3

The population of a certain town grows at a rate proportional to the population. If the population grows from 20 000 to 25 000 in two years, find:

- The population of the town, to the nearest hundred, after a further 8 years. 3
- Calculate the rate of change at this time. 1

(b) i. $\frac{dP}{dt} = kP \quad \therefore P = P_0 e^{kt}$

When $t = 0$, $P = 20\,000$, $\therefore P_0 = 20\,000$

So $P = 20\,000 e^{kt}$

When $t = 2$, $P = 25\,000$

$$25\,000 = 20\,000 e^{2k}$$

$$\frac{5}{4} = e^{2k}$$

$$\ln\left(\frac{5}{4}\right) = 2k$$

$$k = \ln\left(\frac{5}{4}\right) \div 2$$

$$k = 0.111571775$$

When $t = 10$ $P = 20\,000 e^{0.111571775(10)}$
 $= 61\,000$ people (nearest 100)

ii. $\frac{dp}{dt} = 61\,000 \times 0.111571775$

$$= 6\,806 \text{ people / year}$$

4

A block of ice, with mass M grams after t minutes is melting according to the equation $M = M_0 e^{-kt}$.

Initially, there is 100 grams of ice and, after 35 minutes, the block has reduced to 60 grams.

- i) Find the constants M_0 and k (exact form). 3
- ii) Find the rate of melting when the ice weighs 50 grams. 1
- iii) Find how long it takes for the ice to reduce to 5 grams. 2

a) i) $M_0 = 100$
 and $60 = 100 e^{-35k}$
 $e^{-35k} = \frac{3}{5}$

$$-35k = \log\left(\frac{3}{5}\right)$$

$$\underline{k = -\frac{1}{35} \log\left(\frac{3}{5}\right)}$$

ii) $\frac{dM}{dt} = -kM$
 $= \frac{1}{35} \log\left(\frac{3}{5}\right) \times 50$
 $\approx 0.73 \text{ g/min}$

iii) $5 = 100 e^{-kt}$

$$\frac{1}{20} = e^{-kt}$$

$$\therefore -kt = \log\left(\frac{1}{20}\right)$$

$$\text{or } kt = \log 20$$

$$\therefore t = \frac{\log 20}{k}$$

$$\approx 205 \text{ min.}$$

5

A particle moves so that its position x cm from a fixed point 0 after t seconds, is given by $x = 2te^{-t}$.

- i) Find the particle's initial position. 1
- ii) Show that $v = 2e^{-t}(1 - t)$. 1
- iii) Find when the particle is stationary. 1
- iv) Describe the particle's position as $t \rightarrow \infty$ 1
- v) Use the above information to sketch, on coordinate axes, a graph of the particle's position over time.
Show on your curve the time where you expect the acceleration to be zero.
Justify the location. 2

b-) i) $t=0, x = 0 \times e^0$
 $x = 0$

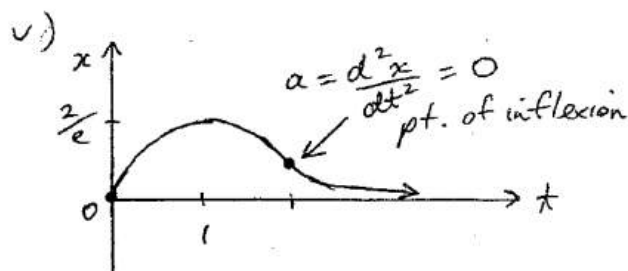
ii) $v = \frac{dx}{dt}$

$$\begin{aligned} &= 2 \times e^{-t} + e^{-t} \times (-1) \times 2t \\ &= 2e^{-t} - 2te^{-t} \\ &= 2e^{-t}(1-t) \end{aligned}$$

iii) $v = 0 \Rightarrow \underline{t = 1 \text{ second}}$

iv) $x = \frac{2t}{e^t}$

as $t \rightarrow \infty, \underline{x \rightarrow 0^+}$



6

A particle is moving along the x -axis. Its position at time t is given by $x = 5e^{-t} \sin t$.

- (i) Show that its velocity is given by $v = 5e^{-t}(\cos t - \sin t)$. 1
- (ii) Where is the particle initially, and what is its initial velocity? 2
- (iii) At what time during the interval $0 \leq t \leq \pi$ is the particle stationary? 2
- (iv) Assuming that its acceleration at time t is $\ddot{x} = -10e^{-t} \cos t$, find the time during the interval $0 \leq t \leq \pi$ when the acceleration is zero. 1

$$(c) x = 5e^{-t} \sin t$$

$$(i) \frac{dx}{dt} = -5e^{-t} \sin t + 5e^{-t} \cos t \checkmark$$

$$= 5e^{-t}(\cos t - \sin t)$$

$$(ii) t=0, x=0, v=5 \quad \checkmark \checkmark$$

$$(iii) \frac{dx}{dt} = 0 \text{ when } \cos t - \sin t = 0$$

$$\frac{\sin t}{\cos t} = 1 \quad \checkmark$$

$$\tan t = 1$$

$$\therefore t = \frac{\pi}{4} \quad \checkmark$$

$$(iv) \ddot{x} = -10e^{-t} \cos t$$

acceleration is zero when $\cos t = 0$

$$\text{i.e. } t = \frac{\pi}{2} \quad \checkmark$$

7

Sophie has a toy that she uses to blow spherical bubbles. The rate of change of the volume $V \text{ cm}^3$ of a bubble is given by

$$\frac{dV}{dt} = \frac{6t}{t^2 + 1} \text{ cm}^3/\text{s}.$$

- (i) Find the equation for the volume V of a bubble t seconds after Sophie starts blowing. Assume that the initial volume of a bubble is zero. 2
- (ii) A bubble will burst when its radius exceeds 1.5 cm. Sophie takes a deep breath and blows a bubble. After how many seconds of blowing will it burst? Give your answer correct to one decimal place. 3

$$b)(i) \frac{dV}{dt} = \frac{6t}{t^2 + 1}$$

$$V = 3 \int \frac{2t}{t^2 + 1} dt$$

$$= 3 \ln(t^2 + 1) + C \quad \checkmark$$

$$t=0 \quad V=0 \quad C=0$$

$$\therefore V = 3 \ln(t^2 + 1) \quad \checkmark$$

$$(ii) V = \frac{4}{3} \pi r^3, \quad r = 1.5$$

$$= \frac{4}{3} \times \pi \times \left(\frac{3}{2}\right)^3$$

$$= \frac{9\pi}{2} \quad \checkmark$$

$$\frac{9\pi}{2} = 3 \ln(t^2 + 1) \quad \checkmark$$

$$\ln(t^2 + 1) = \frac{3\pi}{2}$$

$$t^2 + 1 = e^{\frac{3\pi}{2}}$$

$$t^2 = e^{\frac{3\pi}{2}} - 1$$

$$t \approx 10.5 \text{ s} \quad \checkmark$$

8

The population P of mosquitoes in a laundry is growing exponentially according to the equation $P = 50e^{kt}$, where t is the time in days after the insects are first counted. After four days the population has doubled.

- Find the exact value of the constant k .
- How many mosquitoes will there be after 10 days?
- At what rate is the population increasing after 10 days?

2

2

1

(a) $P = 50e^{kt}$

(i) when $t=4$, $P=100$.

$$100 = 50e^{4k} \checkmark$$

$$2 = e^{4k}$$

$$4k = \ln 2$$

$$k = \frac{1}{4} \ln 2 \checkmark$$

(ii) put $t=10$.

$$P = 50e^{\frac{10}{4} \ln 2}$$

$$P = 50e^{\frac{5}{2} \ln 2} \checkmark$$

$$P = 50(e^{\ln 2})^{\frac{5}{2}}$$

$$P = 50 \times 2^{\frac{5}{2}}$$

$$P = 282.8 \quad (200\sqrt{2})$$

approximately 283 $\checkmark$
mosquitoes.

(iii) $\frac{dP}{dt} = kP$

$$= \frac{1}{4} \ln 2 \times 282.8$$

$$\frac{dP}{dt} \doteq 49 \text{ mosquitoes/day}$$

$$\checkmark (50\sqrt{2} \ln 2)$$

9

A swimming pool is being emptied. The volume of water L litres in the pool after t minutes is given by the equation

$$L = 1000(20 - t)^3.$$

(i) Find the rate at which the pool is emptying after 10 minutes.

2

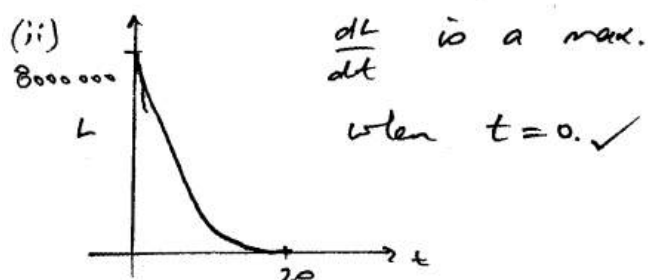
(ii) When is the pool emptying at a maximum rate?

1

(a) $L = 1000(20 - t)^3$
 (i) $\frac{dL}{dt} = -3000(20 - t)^2$ ✓

at $t = 10$. $\frac{dL}{dt} = -300000$

After 10 minutes the pool is being emptied at 300 000 L/minute. ✓



The pool is being emptied at a maximum rate initially.

10

A particle moves in a straight line so that after t seconds ($t \geq 0$) its velocity v is given by $v = \left(\frac{2}{1+t} - t \right)$ m/s. The displacement of the particle from the origin is given by x metres.

- (i) Find the acceleration of the particle when $t = 0$. 2
- (ii) If the particle is initially at the origin, find the displacement as a function of t . 2
- (iii) When is the particle stationary? 1
- (iv) How far does the particle travel in the first 2 seconds? Give your answer correct to three significant figures. 1

(c)

$$v = \frac{2}{1+t} - t \quad \text{m/s}$$

$$(i) \quad \frac{dv}{dt} = \frac{-2}{(1+t)^2} - 1$$

$$a = \frac{-2}{(1+t)^2} - 1 \quad \checkmark$$

when $t = 0$.

$$a = -2 - 1$$

$$a = -3 \text{ m/s}^2 \quad \checkmark$$

$$(ii) \quad \frac{dx}{dt} = \frac{2}{1+t} - t$$

$$x = 2 \ln(1+t) - \frac{t^2}{2} + c \quad \checkmark$$

when $t=0$, $x=0$, $\therefore c=0$

$$x = 2 \ln(1+t) - \frac{t^2}{2} \quad \checkmark$$

$$(iii) \quad v=0. \quad \frac{2}{1+t} - t = 0$$

$$2 - t(1+t) = 0$$

$$2 - t - t^2 = 0$$

$$t^2 + t - 2 = 0$$

$$(t-1)(t+2) = 0 \quad \checkmark$$

$$t = 1 \quad \text{or} \quad t = -2 \quad (\text{omit}) \quad \checkmark$$

The particle is stationary at $t=1$ s

$$(v) \quad \begin{array}{ccccc} x & 0 & 0.8863 & 0.1972 & \dots \\ t & 0 & 1 & 2 & \dots \end{array}$$

$$x = 2 \ln(1+t) - \frac{t^2}{2}$$

distance travelled =

$$0.8863 + (0.8863 - 0.1972)$$

$$= 1.5754$$

$$= 1.58 \text{ m (3 sig. fig.)} \quad \checkmark$$

11

A particle is moving on the x -axis. It starts from the origin O , and at time t seconds its velocity $v \text{ ms}^{-1}$ is given by $v = 1 - 2 \sin t$. Let $t = t_1$ and $t = t_2$ be the first two times that the particle comes to rest.

- Find t_1 and t_2 .
- Sketch the velocity function for $0 \leq t \leq 2\pi$.
- Find the acceleration at $t = t_1$ and $t = t_2$.
- Find the displacement function.
- Hence, or otherwise, find the distance travelled between $t = t_1$ and $t = t_2$.

2

2

2

2

1

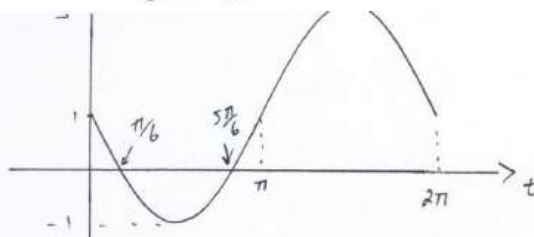
(i) $v = 1 - 2 \sin t$

$0 = 1 - 2 \sin t$

$\frac{1}{2} = \sin t$

$\therefore t = \frac{\pi}{6}, \frac{5\pi}{6}$

(ii)



(iii)

$a = \frac{dv}{dt}$

$= -2 \cos t$

At $t_1 = \frac{\pi}{6}$, $a = -2 \cos \frac{\pi}{6}$
 $= -\sqrt{3} \text{ ms}^{-2}$

$t_2 = \frac{5\pi}{6}$, $a = -2 \cos \frac{5\pi}{6}$
 $= \sqrt{3} \text{ ms}^{-2}$

(iv)

$x = \int v dt$
 $= t + 2 \cos t + c$

$t = 0, x = 0$

$0 = 0 + 2 + c$

$\therefore c = -2$

So $x = t + 2 \cos t - 2$

(v)

$t = \frac{\pi}{6}$, $x = \frac{\pi}{6} + \sqrt{3} - 2$

$t = \frac{5\pi}{6}$, $x = \frac{5\pi}{6} - \sqrt{3} - 2$

So distance = $\left| \left(\frac{5\pi}{6} - \sqrt{3} - 2 \right) - \left(\frac{\pi}{6} + \sqrt{3} - 2 \right) \right|$
 $= \left| \frac{2\pi}{3} - 2\sqrt{3} \right|$
 $= 2\left(\sqrt{3} - \frac{\pi}{2} \right) \text{ m}$

12

The rate of decay of radium-226 is proportional to the mass M present at that time, so that

$$\frac{dM}{dt} = -kM.$$

Radium-226 has a half-life of 1590 years. That is, the time taken for half the initial mass to decay is 1590 years. A sample of radium-226 begins to decay.

- (i) Show that $M = M_0 e^{-kt}$, where k and M_0 are constants, satisfies the differential equation above. 1
- (ii) Find the value of k . 2
- (iii) How many years will it take for 70% of the sample to decay? 2

(i) $M = M_0 e^{-kt}$

$$\frac{dM}{dt} = -k M_0 e^{-kt}$$

$$= -k M$$

(ii) $M = \frac{M_0}{2}$, $t = 1590$

$$\frac{M_0}{2} = M_0 e^{-k(1590)}$$

$$\frac{1}{2} = e^{-k(1590)}$$

$$-1590k = \ln\left(\frac{1}{2}\right)$$

$$= -\ln 2$$

$$k = \frac{\ln 2}{1590}$$

b) (iii) $M = \frac{3}{10} M_0$

$$\therefore \frac{3}{10} = e^{-kt}$$

$$\ln\left(\frac{3}{10}\right) = -kt$$

$$t = \frac{\ln\left(\frac{3}{10}\right)}{-k}$$

$$= \frac{\ln\left(\frac{10}{3}\right)}{\frac{\ln 2}{1590}}$$

$$= 2761.775 \dots \text{years}$$

$\therefore$ It will take 2762 years.

13

The mass M kg of a radioactive substance present after t years is given by the equation

$$M = M_0 e^{-kt}$$

where k is a positive constant.

After 50 years the substance has been reduced from 20 kg to 10 kg in mass.

- (i) Show that $\frac{dM}{dt} = -kM$. 1
- (ii) State the value of M_0 . 1
- (iii) Find the exact value of k . 2
- (iv) Find the time taken for the substance to lose $\frac{4}{5}$ of its original mass. 2

Answer to the nearest year

i) $M = M_0 e^{-kt}$
 $\frac{dM}{dt} = -k \times M_0 e^{-kt}$ ✓
 $= -kM$

ii) $M_0 = 20 \text{ kg}$ ✓

iii) $t = 50 \text{ yrs}$ $M = 10$, $M_0 = 20$

$$10 = 20 \times e^{-k \times 50}$$

$$\frac{1}{2} = e^{-50k}$$
 ✓

$$\log_e \frac{1}{2} = -50k$$

$$\log_e \frac{1}{2} = k$$
 ✓
 -50

Sign
quest

iv) lose $\frac{4}{5}$ means $\frac{1}{5}$ remains
 $\frac{1}{5} = e^{\frac{\log_e \frac{1}{2}}{50} \times t}$ ✓
 $\log_e \frac{1}{5} = \frac{t \log_e \frac{1}{2}}{50}$

$$\frac{50 \log_e \frac{1}{5}}{\log_e \frac{1}{2}} = t$$

$$t = 116 \text{ years}$$
 ✓

14

A particle P is moving in a straight line so that its velocity v metres per second after t seconds is given by $v = 12 - 4t$.

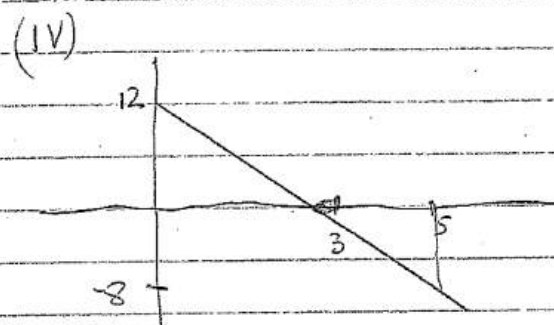
Initially, P is 3 metres to the right of the origin O .

- (i) Find the initial velocity and acceleration of P . 1
- (ii) If the displacement of P from O is x metres, find an expression for x in terms of t . 1
- (iii) Find when and where P is at rest. 1
- (iv) Sketch the graph of $v = 12 - 4t$ for $0 \leq t \leq 5$. 1
- (v) Hence, or otherwise, find the total distance travelled by P during the first 5 seconds. 2

(i) $t = 0$ $v = 12$
 $a = \frac{dv}{dt} =$ -4

(ii) $x = 12t - 2t^2 + C$ to $x = 3$
 $C = 3$
 $x = 12t - 2t^2 + 3$

(iii) $v = 0$ $t = 3$
 $x = (12)(3) - 2(9) + 3$
 $x = 21$



$t = 0$ $x = 3$
 $t = 3$ $x = 21$ 18 m travelled
 $t = 5$ $x = 13$ 8 -
 total 26 m travelled

or area under vel curve

$\frac{1}{2} \times 12 \times 3 + \frac{1}{2} \times 2 \times 8$

$= 26 \text{ m}$

15

Two cultures of bacteria are prepared in a laboratory. They are to be used to test the effectiveness of two drugs. One culture has 1000 bacteria and Drug A reduces this number to 250 in 5 minutes. The other culture has 1250 bacteria and Drug B reduces this number to 500 in 3 minutes. Both cultures are reduced according to the model $N = N_0 e^{-kt}$ where N is the number of bacteria and t is the time since the drug was administered in minutes.

- (i) Find the value for k of both cultures. (2)
- (ii) Which drug is more effective in reducing the number of bacteria?
State a brief reason to support your answer. (1)
- (iii) How long will it take for Drug B to reduce the second culture to 10% of its original number of bacteria? (2)

(b)i)	$\begin{aligned} \text{A: } 250 &= 1000e^{-5k} \\ 0.25 &= e^{-5k} \\ \ln(0.25) &= -5k \\ k &= \frac{\ln(0.25)}{-5} \\ k &= 0.277.... \end{aligned}$	$\begin{aligned} \text{B: } 500 &= 1250e^{-3k} \\ 0.4 &= e^{-3k} \\ \ln(0.4) &= -3k \\ k &= \frac{\ln(0.4)}{-3} \\ k &= 0.305.... \end{aligned}$
ii)	The 'k' decay constant for B is greater than the one for A	
iii)	$\begin{aligned} \text{B: } 125 &= 1250 e^{-0.305t} \\ 0.1 &= e^{-0.305t} \\ \ln(0.1) &= -0.305t \\ t &= \frac{\ln(0.1)}{-0.305} = 7.53... \text{minutes} \end{aligned}$	

16

A ball falls from rest in a fluid medium with its acceleration given by

$$\ddot{x} = 10e^{-\frac{1}{3}t} \text{ cm/sec}^2$$

where x is the distance in centimetres below the origin at time t seconds.

- (i) Find the velocity-time function for the motion of the ball. (2)
- (ii) What is the limiting velocity of the ball. (1)
- (iii) How far does the ball travel in the first 3 seconds? (2)
- (iv) A second ball moves with the same acceleration as the first ball but it is initially forced upwards in the fluid with a speed of 10 cm/sec from a point 2 metres below the first ball at the same instant as the first ball is dropped. Do the balls ever collide and, if so, when? (4)

(i) Velocity = $\int \ddot{x} dt$

$$\begin{aligned}\dot{x} &= \int 10e^{-\frac{1}{3}t} dt \\ &= 10 \int e^{-\frac{1}{3}t} dt \\ &= 10 \times -3 \int \frac{1}{3} e^{-\frac{1}{3}t} dt \\ &= -30e^{-\frac{1}{3}t} + k\end{aligned}$$

$$\begin{aligned}\text{at } t=0, \dot{x} &= 0 \\ 0 &= -30e^{-\frac{1}{3} \times 0} + k \\ 0 &= -30 + k \\ 30 &= k\end{aligned}$$

$$\text{Velocity} = -30e^{-\frac{1}{3}t} + 30$$

(ii) Limiting velocity at $t = \infty$

$$\begin{aligned}\text{as } t \rightarrow \infty \\ \dot{x} &= -30e^{-\frac{1}{3}t} + 30 \\ &= -30e^{-\infty} + 30 \\ &= 30 \text{ cm/sec}\end{aligned}$$

(iii) Travel in first 3 seconds therefore need to find distance/ time function

$$\begin{aligned}\dot{x} &= -30e^{-\frac{1}{3}t} + 30 \\ x &= \int -30e^{-\frac{1}{3}t} + 30 dt \\ &= 90e^{-\frac{1}{3}t} + 30t + k_2 \\ \text{At } t=0, d &= 0 \\ 0 &= 90 + k_2 \\ -90 &= k_2 \\ x &= 90e^{-\frac{1}{3}t} + 30t - 90 \\ \text{At } t=3 \\ x &= 90e^{-1} + 90 - 90 \\ &= 33.1 \text{ cm}\end{aligned}$$

(iv) Ball Two

$$\text{at } t=0, v = -10$$

$$\begin{aligned}-10 &= -30e^{-\frac{1}{3} \times 0} + k_3 \\ 20 &= k_3\end{aligned}$$

$$V_2 = -30e^{-\frac{1}{3}t} + 20$$

$$\dot{x}_{\text{Ball 2}} = -30e^{-\frac{1}{3}t} + 20$$

$$\begin{aligned}x_{\text{Ball 2}} &= \int -30e^{-\frac{1}{3}t} + 20 dt \\ &= 90e^{-\frac{1}{3}t} + 20t + k_2\end{aligned}$$

$$\text{At } t=0, d = 200$$

$$200 = 90 + k_2$$

$$110 = k_2$$

$$x_{\text{Ball 2}} = 90e^{-\frac{1}{3}t} + 20t + 110$$

Balls will collide when they are at the same distance ie:

$$x_{\text{Ball 1}} = x_{\text{Ball 2}}$$

$$90e^{-\frac{1}{3}t} + 30t - 90 = 90e^{-\frac{1}{3}t} + 20t + 110$$

$$10t = 200$$

$$t = 20 \text{ seconds}$$

therefore they will collide after 20 seconds

17

A particle P moves such that the displacement x cm from 0 after t seconds is given by $x = t^3 - 3t$, $t \geq 0$.

- (i) Find the initial velocity. 2
- (ii) Find when the particle is at rest. 2
- (iii) Is the particle speeding up or slowing down at $t = 2$? Give reasons. 2

cont'd b) i) $x = t^3 - 3t$

$$\dot{x} = 3t^2 - 3$$

$$t = 0 \quad \dot{x} = 3(0)^2 - 3$$

$$= -3 \text{ cm/s}$$

$\therefore$ moving to left at 3 cm/s

ii) rest when $\dot{x} = 0 = 3t^2 - 3$

$$3t^2 = 3$$

$$t^2 = 1$$

$$t = \pm 1 \text{ but } t > 0$$

$\therefore t = 1$ particle is at rest

iii) $t = 2 \quad \dot{x} = 3(2)^2 - 3 \quad \ddot{x} = 6t$
 $= 9 \quad = 6(2)$

$$\text{vel} > 0$$

$$\text{accel} = 12 > 0$$

Speeding up since v and a are same sign.

18

Spherical mothballs, each of mass M milligrams, slowly vaporise according to the equation $\frac{dM}{dt} = -4\pi(10-t)^2$, where t is in days. When they are manufactured, they have mass $\frac{4000\pi}{3}$ milligrams.

(i) Find after how many days a mothball ceases to vaporise.

1

(ii) Find M as a function of t .

2

(iii) Find the mass of a mothball after 7 days.

1

$$a) \frac{dM}{dt} = -4\pi(10-t)^2$$

i) Vaporising when $\frac{dM}{dt} = 0 = -4\pi(10-t)^2$
 ceases $\therefore t = 10 \text{ days}$ (1)

$$ii) M = \int -4\pi(10-t)^2 dt$$

$$M = \frac{-4\pi(10-t)^3}{3 \times -1} + C \quad \checkmark$$

$$M = +\frac{4\pi}{3}(10-t)^3 + C$$

$$t=0 \quad M = \frac{4000\pi}{3}$$

$$\frac{4000\pi}{3} = \frac{4\pi}{3}(10-0)^3 + C \quad (2)$$

$$\frac{4000\pi}{3} = \frac{4000\pi}{3} + C$$

$$C = 0$$

$$\therefore M = \frac{4\pi}{3}(10-t)^3 \quad \checkmark$$

$$iii) t=7 \quad M = \frac{4\pi}{3}(10-7)^3$$

$$= \frac{4\pi}{3} \times 27 \quad (1)$$

$$= 36\pi \text{ mg.}$$

19

The population N of native birds on an island after t years is given by the equation $N = N_0 e^{0.03t}$, where N_0 is a constant. There are 8000 native birds on January 1, 2007 (when $t = 0$).

(i) Show that $N_0 = 8000$.

1

(ii) Find the rate at which the population is growing on January 1, 2007.

2

A migrant bird population M settles on the island on January 1, 2007 and begins to compete with the native bird population. The number of migrant birds M after t years is given by $M = 200e^{kt}$, where k is a constant. After 10 years the migrant population has grown to 1000.

(iii) Find the value of the growth constant k , and show $k \approx 0.1609$.

2

(iv) Find the number of years it takes for the migrant population to overtake the native population.

3

b) i) $N = N_0 e^{0.03t}$
 $t = 0 \quad N = 8000 = N_0 e^{0.03 \times 0}$
 $8000 = N_0 \times e^0$ (1)
 $\therefore N_0 = 8000 \therefore N = 8000e^{0.03t}$
 ii) $\frac{dN}{dt} = 0.03 \times 8000e^{0.03t}$ ✓ (2)
 $t = 0 \quad \frac{dN}{dt} = 0.03 \times 8000e^0 = 240 \text{ b/y}$

b) ii) $M = 200e^{kt}$
 $t = 10 \quad M = 1000 = 200e^{k \times 10}$
 $\frac{1000}{200} = e^{10k}$
 $5 = e^{10k}$ (2)
 $\ln 5 = 10k$
 $k = \frac{1}{10} \ln 5$
 $\approx 0.16094...$
 iv) $t = ?$ when $M \geq N$?
 $200e^{0.1609t} \geq 8000e^{0.03t}$
 $e^{0.1609t} \geq 40e^{0.03t}$
 $\frac{e^{0.1609t}}{e^{0.03t}} \geq 40$
 $e^{0.1609t - 0.03t} \geq 40$
 $e^{0.1309t} \geq 40$ ✓
 $0.1309t \geq \ln 40$ (3)
 $\therefore t \geq \frac{\ln 40}{0.1309}$
 $\therefore t \geq 28.18...$ ✓
 $\therefore$ It takes about 29 years.

20

The mass M in grams of a radioactive substance may be expressed as $M = Ae^{-kt}$ where t is the time in years and k is a constant.

- (i) If initially the mass is 10 grams, find A . 1
- (ii) After 5 years the mass is 9 grams. Find the mass after 20 years. 2

(b) (i) $M = Ae^{-kt}$
 $t = 0 \quad M = 10 \quad 10 = Ae^0$
 $\therefore A = 10$

(ii) $t = 5 \quad M = 9$
 $9 = 10e^{-k \times 5}$
 $\frac{9}{10} = e^{-5k}$
 $\ln \frac{9}{10} = -5k$
 $k = -\frac{1}{5} \ln \left(\frac{9}{10} \right) \div 0.0210721$
 $\text{When } t = 20 \quad M = 10e^{-20 \times 0.0210721}$
 $= 6.561$
mass is 6.561 grams

21

A rainwater tank with a volume of 9m^3 is installed in a new house. At 8am rain begins to fall and flows into the empty tank at the rate given by

$$\frac{dV}{dt} = \frac{36t}{t^2 + 20}$$

where t is the time in hours and V is the volume measured in cubic metres ($t = 0$ is represented by 8am).

- (i) Show that the volume of water in the tank at time t is given by

$$V = 18 \log_e \left(\frac{t^2 + 20}{20} \right), \quad t > 0.$$

2

- (ii) Find the time when the tank will be completely filled with water (to the nearest minute). 3

- (iii) Later, when the tank is full and the rain has stopped, Louise turns on the pump which pumps the water out at the rate given by $\frac{dV}{dt} = \frac{t^2}{k}$. The pump continues for 5 hours until the tank is empty. Find the value of k . 3

$$\begin{aligned} \text{(b)(i)} \quad \frac{dV}{dt} &= \frac{36t}{t^2 + 20} \\ V &= \int \frac{36t}{t^2 + 20} dt \\ &= 18 \int \frac{2t}{t^2 + 20} dt \\ V &= 18 \log_e (t^2 + 20) + C \\ t=0 \quad V=0 \quad 0 &= 18 \log_e (0 + 20) + C \\ C &= -18 \log_e (20) \\ V &= 18 \log_e (t^2 + 20) - 18 \log_e 20 \\ &= 18 \log_e \left(\frac{t^2 + 20}{20} \right) \end{aligned}$$

- (ii) Tank will be filled when $V=9$

$$\begin{aligned} 9 &= 18 \log_e \left(\frac{t^2 + 20}{20} \right) \\ \frac{1}{2} &= \log_e \left(\frac{t^2 + 20}{20} \right) \\ \frac{t^2 + 20}{20} &= e^{\frac{1}{2}} \\ t^2 + 20 &= 20e^{\frac{1}{2}} - 20 \\ t^2 &= 12.9744 \\ t &= \sqrt{12.9744} \\ &\quad \text{take +ve ft} \\ &\quad \text{time} \end{aligned}$$

$$\begin{aligned} t &= 3.6 \text{ hours} \\ &= 3 \text{ hrs } 36 \text{ mins} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad V &= \int \frac{t^2}{k} dt \\ V &= \frac{t^3}{3k} + C \\ t=0 \quad V=9 \quad 9 &= 0 + C \\ V &= \frac{t^3}{3k} + 9 \\ \text{But when } t=5 \quad V &= 0 \\ 0 &= \frac{5^3}{3k} + 9 \\ -9 &= \frac{125}{3k} \\ k &= -\frac{125}{27} \end{aligned}$$

$$\begin{aligned} \text{or} \\ V &= \int_0^5 \frac{t^2}{k} dt \\ 9 &= \left[\frac{t^3}{3k} \right]_0^5 \\ 9 &= \left(\frac{125}{3k} - 0 \right) \\ 3k &= \frac{125}{9} \\ k &= \frac{125}{27} \\ &\quad \text{emptying some} \\ &\quad -125/9 \end{aligned}$$

22

A particle moves along the x axis with acceleration at time t given by

$$\frac{d^2x}{dt^2} = 12 \cos(2t)$$

Find x in terms of t given that when $t = 0$, $x = 0$ and $\frac{dx}{dt} = 0$

(3)

$$\ddot{x} = 12 \cos 2t$$

$$\rightarrow \dot{x} = 6 \sin 2t + c$$

$$\text{when } t = 0, \dot{x} = 0 \therefore c = 0$$

$$\therefore \dot{x} = 6 \sin 2t$$

$$x = -3 \cos 2t + c$$

$$\text{when } t = 0, x = 0 \therefore c = 3$$

$$\therefore x = 3 - 3 \cos 2t$$

23

A tank is emptied by a tap from which the water flows so that, until the flow ceases, the rate of flow after t minutes is R litres/minute where $R = -(t - 6)^2$

- (i) What is the initial rate of flow? (1)
- (ii) How long does it take to empty the tank? (1)
- (iii) How long will it take (to the nearest second) for the flow to drop to 20 litres/minute (2)
- (iv) How much water was in the tank initially? (2)

a) i) $R = -(t - 6)^2$

when $t = 0$, $R = -36$ l/minute.

ie Water is flowing out at 36 l/minute

(ii) when $R = 0$, $0 = -(t - 6)^2$

$t = 6$ seconds

(iii) $R = -20$, $-20 = -(t - 6)^2$

$20 = (t - 6)^2$

$t - 6 = \pm \sqrt{20}$

$t = 6 \pm \sqrt{20}$

But $t \leq 6$, $\therefore t = 6 - \sqrt{20}$ seconds

(iv) $\frac{dV}{dt} = -(t - 6)^2$

$V = -\frac{(t - 6)^3}{3} + C$

when $t = 6$, $V = 0 \rightarrow C = 0$.

$\therefore V = -\frac{(t - 6)^3}{3}$

when $t = 0$, $V = -\frac{(-6)^3}{3}$

$= 72$ litres

24

After its engine is cut, a biplane travels in a straight line with negative acceleration proportional to its

velocity (V) i.e. $\frac{dV}{dt} = -kV$

The plane is travelling at 400m/s when the engine is cut and 10 seconds later the velocity is 250m/s.

- (i) Show that this equation is satisfied by the equation $V = V_0 e^{-kt}$ (1)
- (ii) Find the value of V_0 and k (to 3 decimal places) (3)
- (iii) Find how long it takes for the velocity to fall to 100m/s (answer to 1 decimal place) (2)

b) $\frac{dV}{dt} = -kV$

i) $V = V_0 e^{-kt}$

$$\frac{dV}{dt} = -k V_0 e^{-kt}$$

$$= -kV \text{ as required.}$$

(ii) $400 = V_0 e^0 \rightarrow V_0 = 400.$

$$\therefore 250 = 400 e^{-10k} \text{ when } t=10$$

$$0.625 = e^{-10k}$$

$$-10k = \ln 0.625$$

$$k = \frac{\ln 0.625}{-10} = 0.047 \text{ (to 3 d.p.)}$$

(iii) $100 = 400 e^{-kt}$

$$0.25 = e^{-kt}$$

$$-kt = \ln 0.25$$

$$t = \frac{\ln 0.25}{-0.047}$$

$$\approx 29.5 \text{ seconds}$$

25

(A) A particle is moving in a straight line. It starts from a fixed point O on the line and at time t seconds its velocity v (ms^{-1}) is given by $v = 3t^2 + 6t - 9$.

- i. Find the initial speed of the particle and its initial direction of motion. 2
- ii. Find when and where the particle is at rest. 2
- iii. Find the position of the particle after 2 seconds. 2
- iv. Find the distance travelled by the particle in the first 2 seconds. 2

i. $v = 3t^2 + 6t - 9$
 $t = 0 \Rightarrow v = -9$

Particle is travelling left at a speed of 9ms^{-1} .

ii. When particle is at rest,

$$v = 0 \Rightarrow 3t^2 + 6t - 9 = 0$$

$$t^2 + 2t - 3 = 0$$

$$(t + 3)(t - 1) = 0$$

$$t > 0 \Rightarrow t = 1$$

iii. $x = t^3 + 3t^2 - 9t$

$$t = 2 \Rightarrow x = 2$$

After 2 seconds the particle is 2m to the right of O .

Particle comes to rest instantaneously after 1 second.

$$v = 3t^2 + 6t - 9$$

$$x = t^3 + 3t^2 - 9t + c$$

$$t = 0, x = 0 \Rightarrow 0 = c$$

$$\therefore x = t^3 + 3t^2 - 9t$$

$$t = 1 \Rightarrow x = -5$$

When particle comes to rest after 1 second, it is 5m to the left of O .

iv. Particle starts at the origin, travels 5m to the left during the 1st second, then travels 7m to the right during the 2nd second. The distance travelled during the first 2 seconds is 12m.

26

A particle moves on a straight line so that its velocity, v m/s, at any time

t seconds is given by $v = (t-1)^4 + \frac{t}{2}$, $t \geq 0$

- (i) Find the initial velocity and show that the particle never stops. 2
- (ii) Find the initial acceleration of the particle. 2
- (iii) Find the least value of the velocity. 2
- (iv) Sketch the velocity-time graph. 2
- (v) Find the distance travelled by the particle in the first 2 seconds. 2

$$(i) \quad t=0, \quad v = (-1)^4 + 0 = 1 \text{ m/s}$$

$$\text{Now, } (t-1)^4 \geq 0 \text{ for all } t \geq 0$$

$$\text{and } \frac{t}{2} > 0 \text{ for all } t > 0$$

$$\therefore v > 0 \text{ for all } t \geq 0 \text{ i.e. particle never stops}$$

$$(ii) \quad \ddot{x} = \frac{dv}{dt} = 4(t-1)^3 + \frac{1}{2}$$

$$\therefore \text{when } t=0, \quad \ddot{x} = -4 + \frac{1}{2} = -3\frac{1}{2} \text{ m/s}^2$$

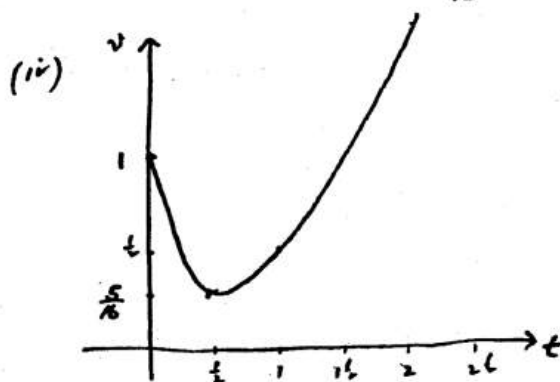
$$(iii) \quad \text{From (i) and (ii), least } v \text{ occurs when } \ddot{x} = 0$$

$$\Rightarrow 4(t-1)^3 + \frac{1}{2} = 0$$

$$\text{or } (t-1)^3 = -\frac{1}{8}$$

$$\therefore t-1 = -\frac{1}{2} \text{ or } t = \frac{1}{2}$$

$$\therefore \text{least } v = \frac{1}{16} + \frac{1}{4} = \frac{5}{16} \text{ m/s}$$



$$(v) \quad x = \int_0^2 \left((t-1)^4 + \frac{t}{2} \right) dt = \left[\frac{(t-1)^5}{5} + \frac{t^2}{4} \right]_0^2$$

$$= \frac{1}{5} + 1 - \left(-\frac{1}{5} + 0 \right) = 1\frac{2}{5} \text{ m}$$

27

In the land of Welf the interest on investments is paid continuously. If an interest rate of $100k\%$ p.a. is given then it can be shown that the amount in the fund after t years, $t \geq 0$, is $A(t)$ where

$A(t) = Pe^{kt}$, P is the initial investment.

- (i) A particular fund, F , in Welf pays 10% p.a. interest. Show that $k = 0.1$ 1
- (ii) Sally invests \$5000 into fund F for 20 years. Find, correct to the nearest dollar, the amount Sally would have after the 20 years. 1
- (iii) Sally wants to be more wealthy in Welf and decides not to terminate her investment of \$5000 until it has grown to at least \$100 000. For how many years will she need to wait? 2
- (iv) Lucy also invests in fund F . She decides to deposit \$1500 into the fund at the start of each year for 20 years. Find, correct to the nearest dollar, the amount Lucy would have after the 20 years. 3

(a) (i) $\therefore 100k = 10 \Rightarrow k = 0.1$

(ii) $A(20) = 5000 e^{0.1 \times 20} = 5000 e^2$
 $= \$36945$, nearest dollar

(iii) Hence, $A(t) = 5000 e^{0.1t} \geq 100000$

$\therefore e^{0.1t} \geq 20$

$\ln e^{0.1t} \geq \ln 20$

$\therefore t \geq \frac{\ln 20}{0.1} = 29.957 \dots$

$\Rightarrow$ Sally needs to wait approx 30 years

(iv) Lucy would have

$$1500 e^{0.1 \times 20} + 1500 e^{0.1 \times 19} + \dots + 1500 e^{0.1 \times 1}$$

$$= 1500 e^1 + 1500 e^2 + 1500 e^3 + \dots + 1500 e^2,$$

is a geometric series where $a = 1500 e^1$
 $r = e^1$

$\therefore$ Lucy would have $\frac{1500 e^1 ((e^1)^{20} - 1)}{e^1 - 1}$

$= \frac{1500 e^1 (e^{20} - 1)}{e^1 - 1}$

$= \$100707$, nearest dollar

28

A tap that empties the water from an urn is turned on slowly so that the volume flow rate, $\frac{dV}{dt}$, varies with time according to the relation $\frac{dV}{dt} = qt^2$, where $t > 0$ and q is a constant. Calculate the total volume of water that flows through the tap in the first 5 seconds if $q = 3.6 \text{ m}^3\text{s}^{-2}$.

$$\frac{dV}{dt} = qt^2 = 3.6t^2$$

$$V = \int_0^5 3.6t^2 dt$$

$$= \left[\frac{3.6t^3}{3} \right]_0^5$$

$$= [1.2t^3]_0^5$$

$$V = 150 \text{ m}^3$$

29

The number of bacteria (N) in a culture is growing exponentially according to the formula $N = 80e^{kt}$.

- | | |
|---|---|
| (i) What is the initial number of bacteria? | 1 |
| (ii) After eight hours the number of bacteria has doubled. Calculate the value of k . Give your answer correct to 3 decimal places. | 2 |
| (iii) How many bacteria, correct to the nearest ten, will there be after 12 hours? | 1 |
| (iv) At what rate will the bacteria be increasing after 12 hours? | 1 |

(a) $N = 80e^{kt}$

(i) when $t=0$, $N = 80e^0$
 $N = 80$

(ii) when $t=8$, $N = 160$.

$\therefore 160 = 80e^{8k}$

$e^{8k} = 2$

$8k = \ln 2$

$k = \frac{\ln 2}{8} = 0.08664$

$\therefore k = 0.087$ (3 d.p.)

(iii) when $t=12$,

$N = 80e^{0.087 \times 12}$

$= 226.27$

$N = 230$ (to nearest ten)

(iv) $dN/dt = k \times N$

$k = 0.087 \times 230$

Rate = $19.66/h$ (using exact values)

or $20.016/h$ (using answers from (ii) + (iii))

ignore rounding.

30

The position of a particle moving along the x -axis is given by $x = 2t + e^{-2t}$ where t is the time in seconds and x is measured in centimetres.

- | | |
|---|---|
| (i) Show that the particle is at rest when $t = 0$. | 2 |
| (ii) What is the acceleration of the particle after 1 second? | 2 |
| (iii) What is the limiting velocity of the particle? | 1 |

1) $x = 2t + e^{-2t}$

(i) $\frac{dx}{dt} = 2 - 2e^{-2t}$

when $t=0$, $\frac{dx}{dt} = 2 - 2 \times e^0$
 $= 2 - 2$
 $\boxed{\frac{dx}{dt} = 0}$

$\therefore$ Particle is at rest when $t=0$.

(ii) $\frac{d^2x}{dt^2} = +4e^{-2t}$

when $t=1$, $\boxed{\frac{d^2x}{dt^2} = 4e^{-2} \text{ cm/s}^2}$

(iii) $\lim_{t \rightarrow \infty} \frac{dx}{dt} = \lim_{t \rightarrow \infty} (2 - \frac{2}{e^{2t}})$

$u = 2 - \infty$

$\therefore \boxed{\text{lim vel} = 2 \text{ cm/s}}$

4 = 122x